

Miniseminar on models of curves

Oberseminar, WS 24/25

Let K be a field equipped with a nontrivial (additive) valuation of rank one,

$$v_K : K \rightarrow \mathbb{R} \cup \{\infty\}.$$

We write $\Gamma_K := v_K(K^\times) \subset \mathbb{R}$ for the value group, $\mathcal{O}_K \subset K$ for the valuation ring, $\mathfrak{m}_K \triangleleft \mathcal{O}_K$ for the maximal ideal and $k := \mathcal{O}_K/\mathfrak{m}_K$ for the residue field of v_K .

For most of the talks, we make the following additional assumptions:

- Assumption 1** (a) v_K is discrete, i.e. Γ_K is a discrete subgroup of $\mathbb{R}$,
 (b) K is complete with respect to v_K ,
 (c) k is perfect.

Attention should be paid to where and how exactly these assumptions are used, and to what extent they could possibly be relaxed.

For the applications we have in mind, K would usually be a finite extension of $\mathbb{Q}_p$ or of $\mathbb{Q}_p^{\text{nr}}$. However, for explicit computations it is often useful to work over a number field K_0 which ‘approximates’ the p -adic number field K of interest (this means that K is the completion of K_0 at some p -adic valuation). This approach can be conceptualized by working with the *henselization* of K_0 instead of the completion, and may be a reason to relax the Assumption (b) that K is complete by replacing it with the assumption that it is henselian.

Another reason to be flexible with these assumptions is that it can be useful to work over an algebraically closed base field K , which is, of course, incompatible with assumption (a).

Let X be a *nice* curve over K , i.e. a smooth, projective and absolutely irreducible curve over K . Our main interests lies in constructing and studying integral models of X with respect to the valuation v_K . Here is what we mean by ‘integral model’ exactly.

Definition 1 A model of X is a proper, flat and normal $\mathcal{O}_K$ -scheme $\mathcal{X}$, together with an identification of its generic fiber with X .

If $\mathcal{X}, \mathcal{X}'$ are two models of X , we say that $\mathcal{X}'$ *dominates* $\mathcal{X}$ if there exists a morphism of $\mathcal{O}_K$ -schemes $\mathcal{X}' \rightarrow \mathcal{X}$ which is the identity on the generic fiber X . If such a morphism exists, it is unique and called a *modification* of $\mathcal{X}$. Note that domination equips the set of isomorphism classes of models of X with the structure of a partially ordered set.

Even though there is a lot of ongoing work on finding explicit models with specific properties (for instance, *regular* or *semistable* models), it remains a difficult problem to represent such models in a way which is both general and amenable to explicit computations. Our approach, inspired by nonarchimedean analytic geometry and pioneered by Julian R  th in his thesis ([7]), is to use *valuations*.

The goal of this miniseminar is to get a solid understanding of the two theorems below, both from a theoretical and practical point of view. Before we can state them, we need one more definition.

Definition 2 Let F_X/K denote the function field of X . A valuation $v : F_X \rightarrow \mathbb{R} \cup \{\infty\}$ is called of type II if

- (a) $v|_K = v_K$,
- (b) v is discrete, and
- (c) the extension of residue fields $\kappa(v)/k$ has transcendence degree one.

We write $V_{II}(F_X)$ for the set of all valuations of type II. The multiplicity of $v \in V_{II}(F_X)$ is the index of valuation groups of the extension of valued fields F_X/K :

$$m(v) := (\Gamma_v : \Gamma_K),$$

where $\Gamma_v := v(F_X^\times) \subset \mathbb{R}$ is the value group of v . We say that v is weakly unramified if $m(v) = 1$.

Let $\mathcal{X}$ be a model of X and $Z \subset \mathcal{X}_s$ an irreducible component of its special fiber. As explained in the first talk, we can associate to Z a valuation v_Z of type II. We set

$$V(\mathcal{X}) := \{v_Z \mid Z \subset \mathcal{X}_s \text{ irreducible component}\} \subset V_{II}(F_X).$$

Theorem 1 The map

$$\mathcal{X} \mapsto V(\mathcal{X})$$

defines a monotone bijection between

- the set of isomorphism classes of models of X , and
- the set of finite, nonempty subsets of $V_{II}(F_X)$.

This theorem is the topic of the first two talks.

The purpose of the second theorem is to understand the behaviour of models under base change with respect to an extension of the base field. Usually, L/K will denote a finite field extension. Since v_K is henselian by Assumption 1 (b), it has a unique extension to v_L to L , and then (L, v_L) also satisfies Assumption 1.

Given a model $\mathcal{X}$ of X , we ask ourselves whether the base change $\mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_L$ is again a model of $X_L := X \otimes_K L$. Well, almost: all properties required of a model from Definition 1 hold except possibly normality. We obtain a model of X_L , called the *normalized base change* of $\mathcal{X}$ relative to L/K , by normalizing $\mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_L$.¹

We say that the model $\mathcal{X}$ is *permanent* if $\mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_L$ is normal, for all finite extensions L/K . In other words, if $\mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_L$ is a model. Our second theorem has this to say about permanence:

Theorem 2 (i) *For a model $\mathcal{X}$ of X , the following three statements are equivalent:*

- (a) $\mathcal{X}$ is permanent,
- (b) the special fiber $\mathcal{X}_s$ of $\mathcal{X}$ is reduced,
- (c) all $v \in V(\mathcal{X})$ are weakly ramified over v_K , i.e. v and v_K have the same value group.

(ii) *Given a model $\mathcal{X}$, there exists a finite extension L/K such that the normalized base change of $\mathcal{X}$ relative to L/K is permanent.*

The goal of the third talk is to give a proof of Part (i) in general, and of Part (ii) in the special case where all multiplicities of $\mathcal{X}_s$ are prime to the characteristic of the residue field k . We will not give a proof of the second part in general. In the fourth and fifth talk we will see a proof of Part (ii) in important special cases.

1 Models of curves and valuations

The goal of this talk is to explain all aspects of Definition 1, Definition 2 and the statement of Theorem 1. You should also do some first steps towards the proof of Theorem 1.

In particular:

- Explain that *flat* in Definition 1 is equivalent to $\mathcal{X}$ being integral.
- Explain the definition of v_Z , why it is of type II, and show that $m(v_Z)$ corresponds to the multiplicity of the component Z .

¹Here we use the fact that Assumption 1 implies that $\mathcal{O}_K$ is an *excellent* ring. It follows that $\mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_L$ is an excellent scheme, and therefore its normalization is a finite morphism. See [4, Chapter 8, in particular Theorem 8.2.39 (d)].

- Explain the proofs of Lemmata 3.6 - 3.8 and Proposition 3.9 from [7]. Lemma 3.8 relies on Zariski's Main Theorem, in the form of Corollary 4.4.3 from [4]. You can accept this theorem, but please explain why *normal* is the crucial assumption here.

Illustrate this, as much as possible, with one simple but instructive example.

2 Proof of Theorem 1

In this talk, the proof of Theorem 1 is completed. You can follow [7, §3.2]. The hardest step, Proposition 3.16, is actually [4, Theorem 8.3.36]. Its proof is scattered throughout Liu's book and can only be sketched. If time allows, you can mention or discuss the following points:

- [4, Theorem 8.3.26] actually shows that all models are projective.
- It suffices to assume that $\mathcal{O}_K$ is henselian instead of complete (Assumption 1 (b)), but this assumption cannot be dropped completely.
- There are version of Theorem 1 where Assumption 1 is relaxed even further. See [6], §1.2 and [3].

3 Permanent models and the Epp problem

In this talk you should give a detailed proof of Part (i) of Theorem 2 and an overview of Part (ii).

For Part (i) you can follow [1, §2.2]. The crucial fact from commutative algebra here is Serre's criterion for normality ([4, Theorem 8.2.23]). Instead of a proof it is enough to illustrate this result with a meaningful example.

For Part (ii), first state the theorem of Epp ([2]) and explain why it proves Theorem 2 (ii). Prove Epp's theorem for tamely ramified extensions (this is also called *Abhyankar's Lemma*), and give an explicit example where this proof strategy fails.

Finally, explain that the Semistable Reduction Theorem trivially implies Part (ii) of Theorem 2, but that the converse is not true.

4 MacLane valuations

In his seminal work [5], S. MacLane considered and essentially solved the following problem. Let v_K be a discrete valuation on a field K . What are the extensions of v_K to an (arbitrary) pseudo-valuation v on the polynomial ring $K[x]$? How can we describe such pseudo-valuations explicitly? These results have been applied to the problem of constructing models of curves in J. R  th's thesis, [7]. This talk gives a brief overview.

Let us denote by $V(K[x])$ the set of all pseudovaluations on $K[x]$ extending v_K . Actually, this is the set underlying the K -analytic space $(\mathbb{A}_K^1)^{\text{an}}$ associated to the affine line over K . There is a well-know classification of elements of $V(K[x])$ into four types (type I, II, III and IV). Elements v of type I are the pseudovaluations which are not true valuations, i.e. there is a nonzero polynomial $f \in K[x]$ with $v(f) = \infty$. Elements of type II are the restriction of valuations on $K(x)$ of type II as in Definition 2 to the subring $K[x]$.

Using the results of MacLane we are able to solve the following two problems relevant for us:

- (a) The Epp problem for valuations of type II on $K(x)$. By Theorem 2 this means that if $\mathcal{X}$ is a model of $X := \mathbb{P}_K^1$ for which the corresponding set of valuations $V(\mathcal{X})$ is known, then we can explicitly compute a finite extension L/K such that the normalized base change of $\mathcal{X}$ to L is permanent.
- (b) The general problem of computing the set of extensions of a valuation v on a function field F/K to an arbitrary finite and separable extensions F'/F . The reason for this is that if $F' = F[\alpha]$ and $f \in F[x]$ is the minimal polynomial of α over F , then an extension of v to a valuation on F' is the same thing as an extension of v to a pseudovaluation on $F[x]$ with $v(f) = \infty$.

In this talk you should explain (a) in some details (using results from [5] without proof), and give a brief sketch of (b).

5 p -cyclic extensions

In this last talk we want to apply all of the above to so-called *superelliptic curves of degree p* , i.e. curves Y/K which are generically defined by an equation of the form

$$Y : y^p = f(x),$$

where $f \in K[x]$ is a polynomial and p is equal to the residue characteristic of K .

By design, Y is given as a Kummer cover of the projective line,

$$\phi : Y \rightarrow X := \mathbb{P}_K^1.$$

We will restrict our attention to models $\mathcal{Y}$ of Y which are the normalization of models $\mathcal{X}$ of X . To construct and describe such a model, it then suffices to (a) describe a model $\mathcal{X}$ of X via MacLane valuations on $F_X = K(x)$ and (b) compute the normalization $\mathcal{Y}$ of X by computing the extensions of valuations $v \in V(\mathcal{X})$ to the extension F_Y/F_X . As explained in Talk 4, we know how to do this explicitly.

The main focus of this talk should be on the Epp problem for models $\mathcal{Y}$ as above. For this, consider a valuation $v \in V(\mathcal{X})$. By Talk 4 we know how to solve the Epp problem for v with respect to the extension $F_X = K(x)/K$. We may actually assume that v is the Gauss valuation of the parameter x . The

problem is now to (a) compute the set of all extensions of v to $F_Y = F_X(y) = K(x, y \mid y^p = f(x))$, and (b) to find a finite extension L/K such that v is weakly unramified in the extension F_Y/F_X after we replace K by L .

Assume first that (b) has already been solved, i.e. that v is weakly ramified in F_Y/F_X . We can then find the set of extension of v to F_Y by a certain algorithm which computes so-called *p-approximations* of f , of the form

$$f = h^p + g,$$

and where the valuation $v(g)$ is maximized. In case v has a unique extension to F_Y , we have $0 \leq v(g) \leq p/(p-1)$, and the rational number

$$\delta = p/(p-1) - v(g)$$

is a natural invariant called the *depth Swan conductor* which measures the (fierce) ramification of F_Y/F_X relative to v .

In the second part of the talk you should sketch an approach to solving the Epp problem simultaneously for the whole family of v given by a disk with fixed center and radius varying over an interval. The main idea is to find a p -approximation $f = h^p + g$ which is adapted to all valuations v in the family. To guarantee this condition, one has to choose h and g such that g , written as a polynomial in the parameter x of the family of disks, has p -coefficients of sufficiently high valuation.

References

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